CHAPTER 8

The Appendicular Skeleton

The limbs and the girdles that attach them to the body.

Anatomy & Physiology · Broward-Miami Health Institute

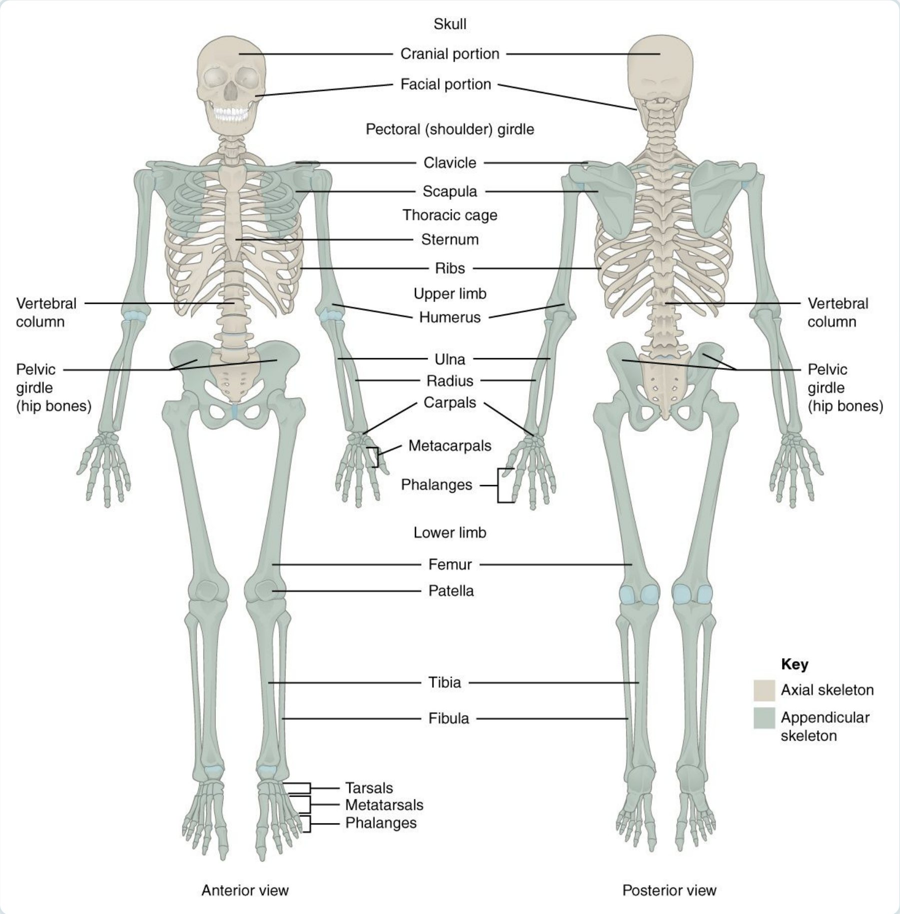
FACULTY EDITION

CHAPTER 8

Learning Objectives

After studying this chapter, students will be able to:

1. Discuss the bones of the pectoral and pelvic girdles, and describe how these unite the limbs with the axial skeleton
2. Describe the bones of the upper limb, including the bones of the arm, forearm, wrist, and hand
3. Identify the features of the pelvis and explain how these differ between the adult male and female pelvis
4. Describe the bones of the lower limb, including the bones of the thigh, leg, ankle, and foot
5. Describe the embryonic formation and growth of the limb bones Your skeleton provides the internal supporting structure of the body



The appendicular skeleton — the limbs and their girdles.

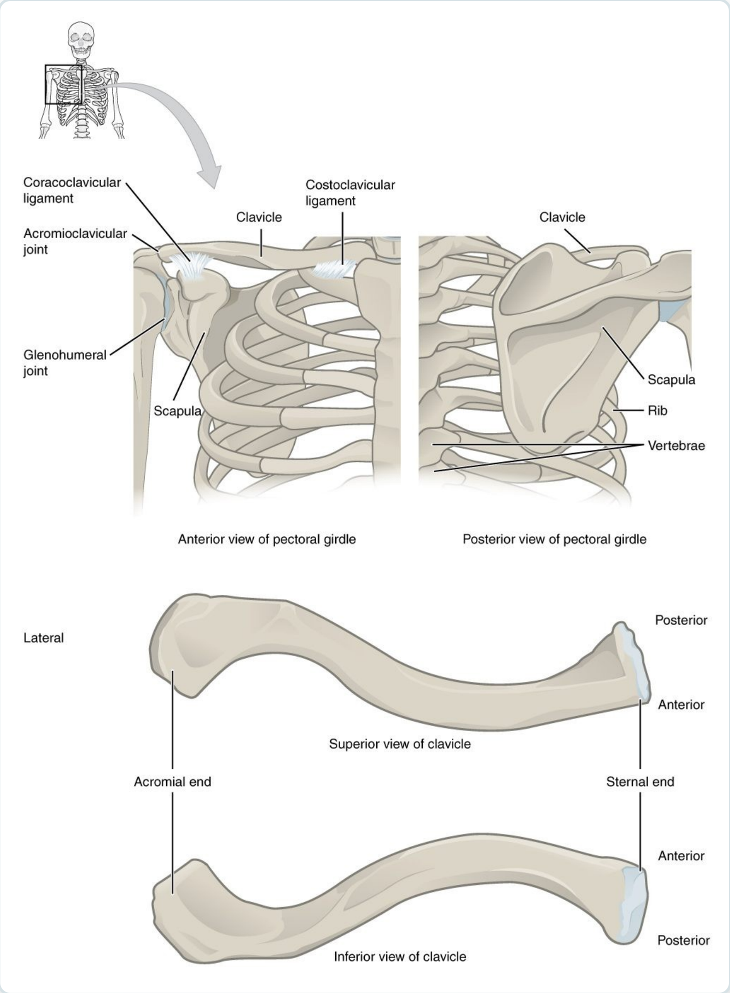
8.1

SECTION

The Pectoral Girdle

The clavicle and scapula that anchor the arm to the body.

The Pectoral Girdle



The pectoral girdle (clavicle and scapula)

- Clavicle and scapula on each side
- Attaches the arm to the axial skeleton
- Highly mobile but less stable
- The clavicle is commonly fractured

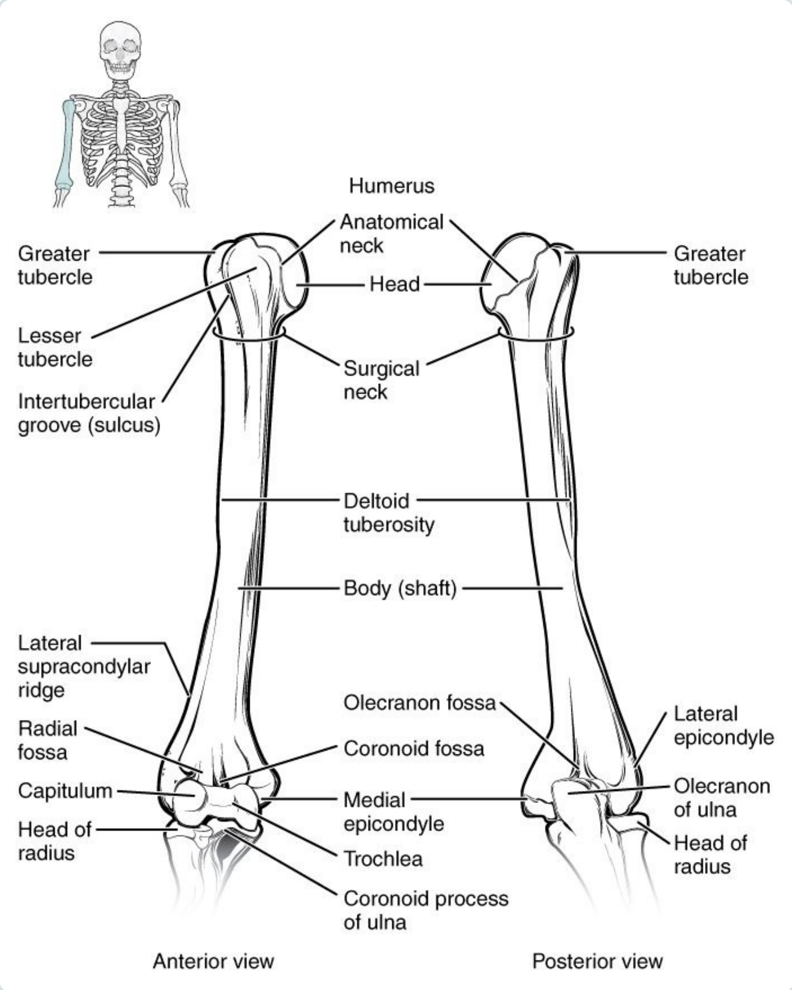
8.2

SECTION

Bones of the Upper Limb

The humerus, radius, ulna, and the bones of the wrist and hand.

The Humerus

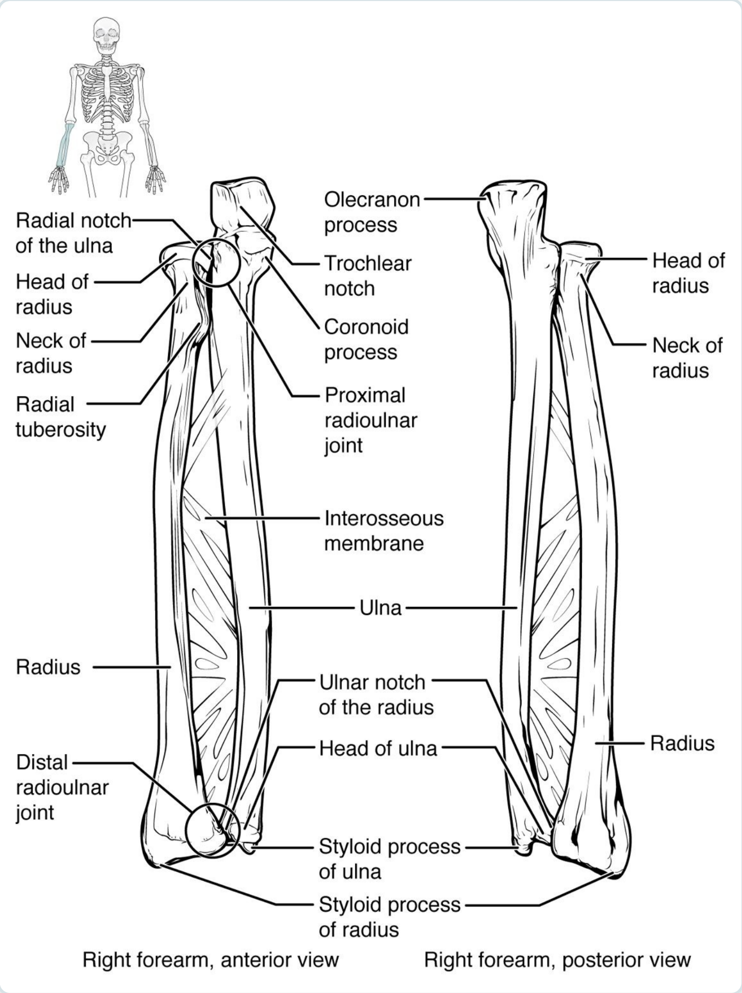


The humerus (arm bone)

- The single bone of the arm
- Head fits the scapula's glenoid cavity
- Articulates with radius and ulna
- Many muscle attachment sites

SECTION 8.2

The Radius and Ulna

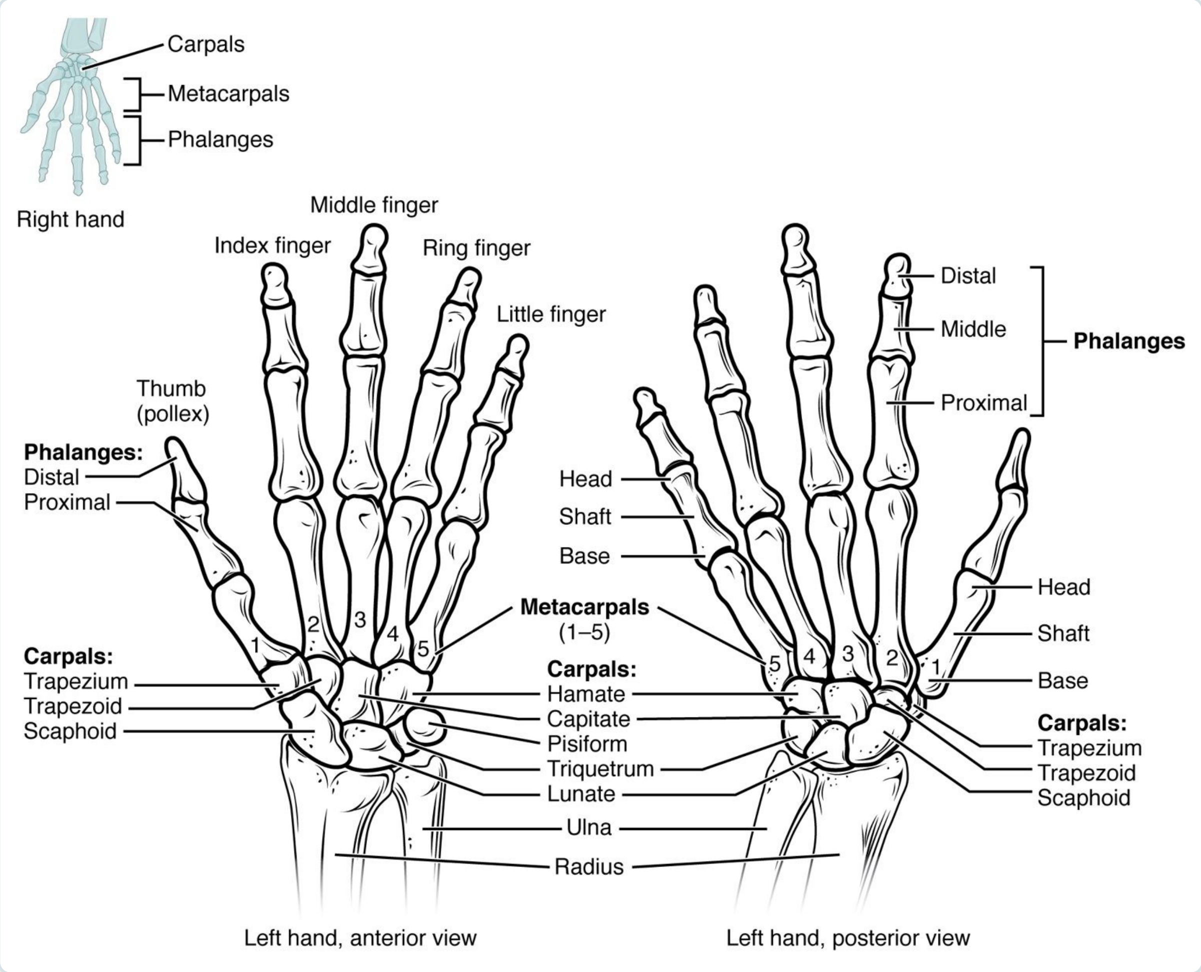


The radius and ulna (forearm)

- The two bones of the forearm
- Ulna is medial (little-finger side)
- Radius is lateral (thumb side)
- They cross during pronation

Bones of the Hand

- 8 carpals form the wrist
- 5 metacarpals form the palm
- 14 phalanges form the fingers
- Built for fine, precise movement

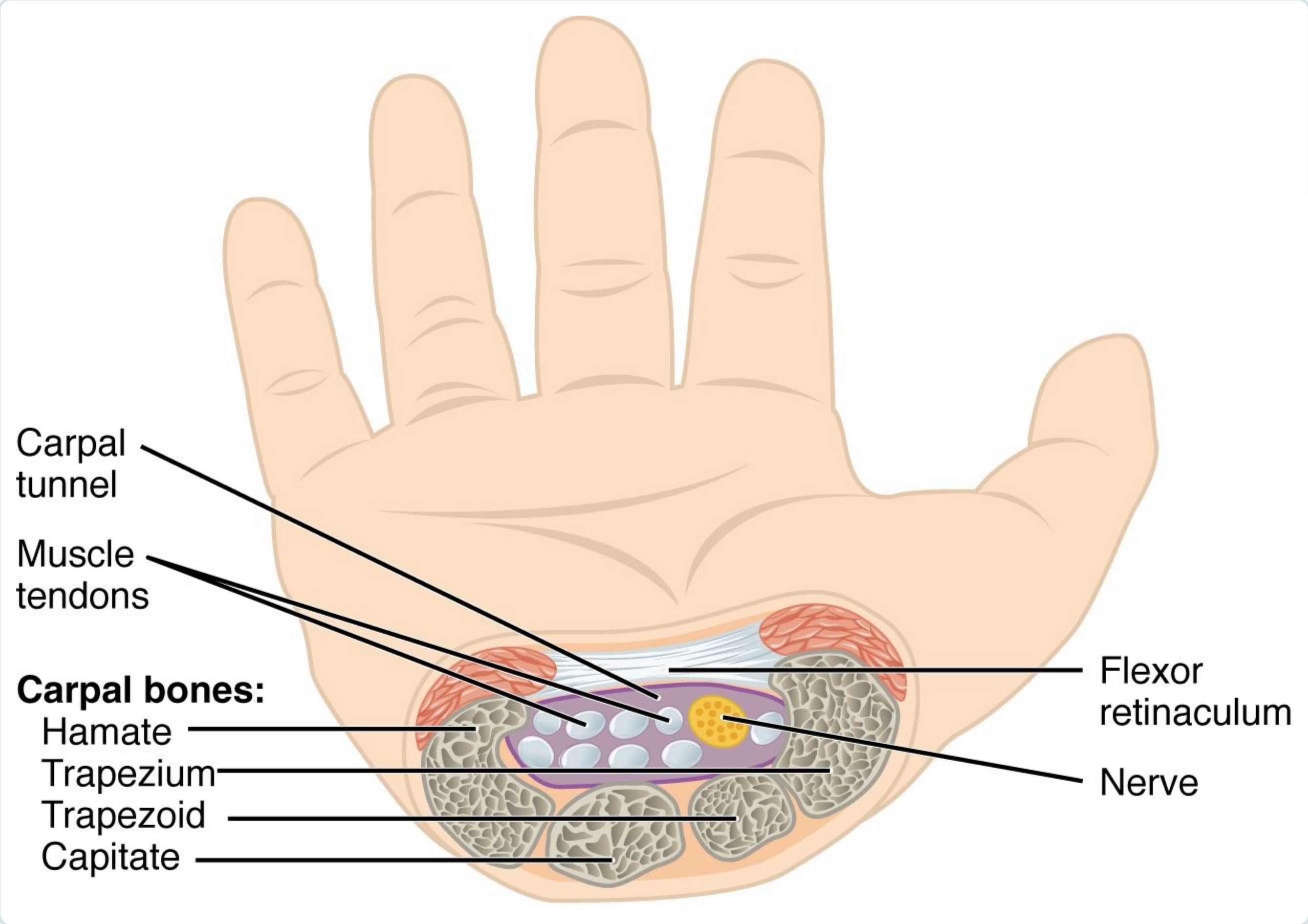


The bones of the wrist and hand

SECTION 8.2

The Carpal Tunnel

- A passage at the front of the wrist
- Formed by carpals + flexor retinaculum
- Carries the median nerve and tendons
- Compression causes carpal tunnel syndrome



The carpal tunnel and median nerve

8.3

SECTION

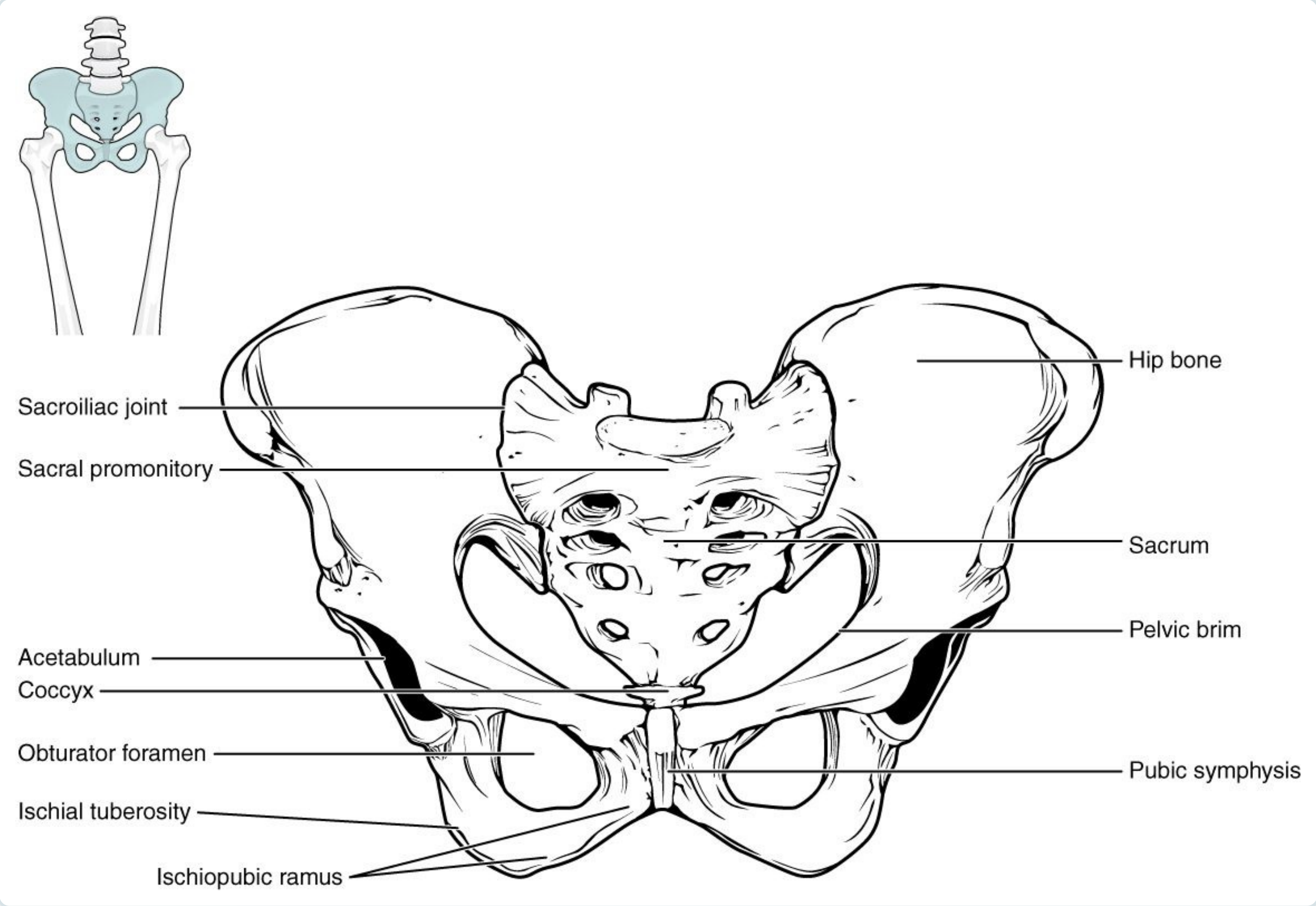
The Pelvic Girdle and Pelvis

The hip bones, the sacrum, and the pelvis they form.

SECTION 8.3

The Pelvis

- Two hip bones + sacrum + coccyx
- Bears the body's weight
- Protects the pelvic organs
- A stable, weight-bearing girdle

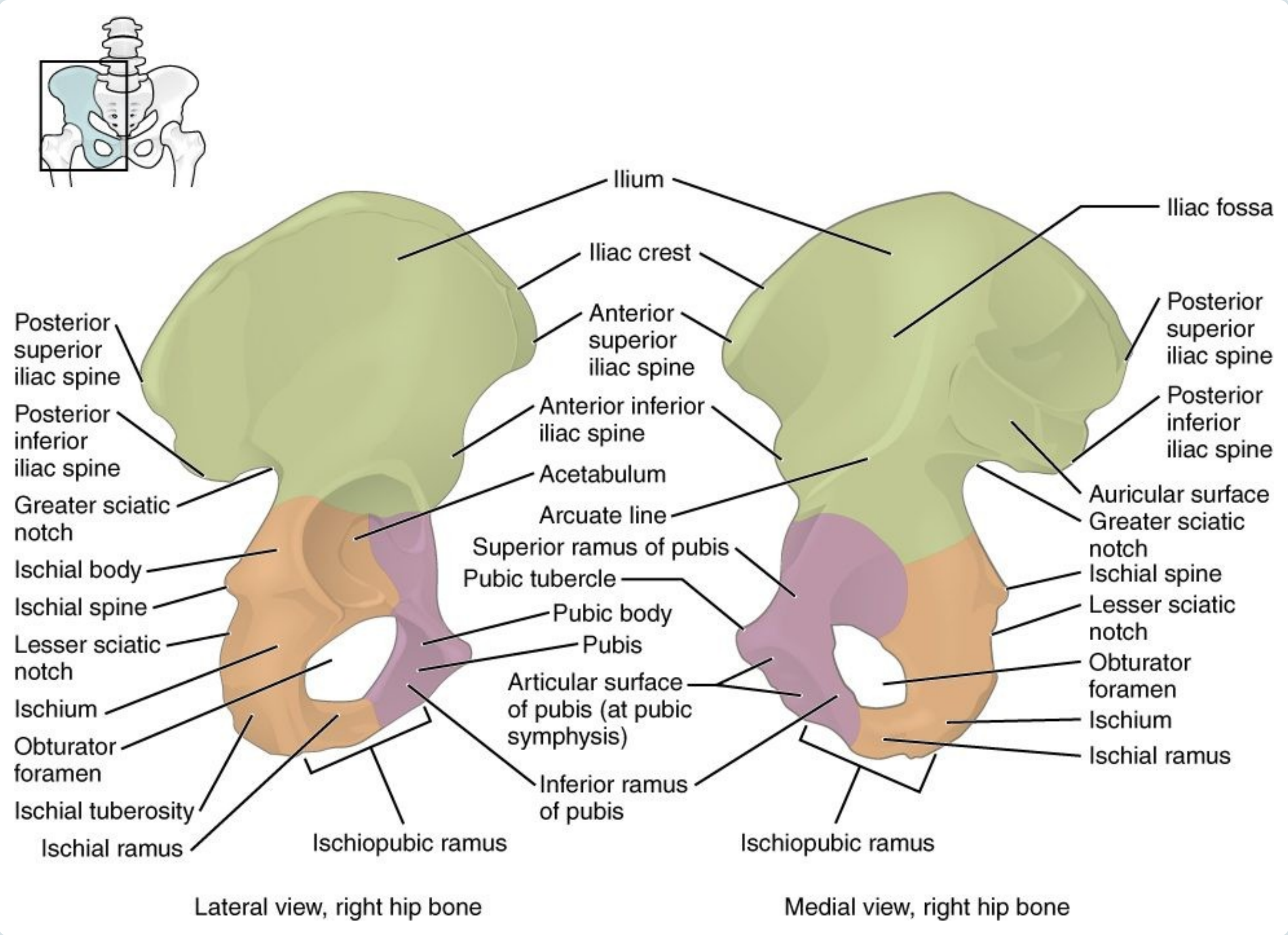


The pelvis

SECTION 8.3

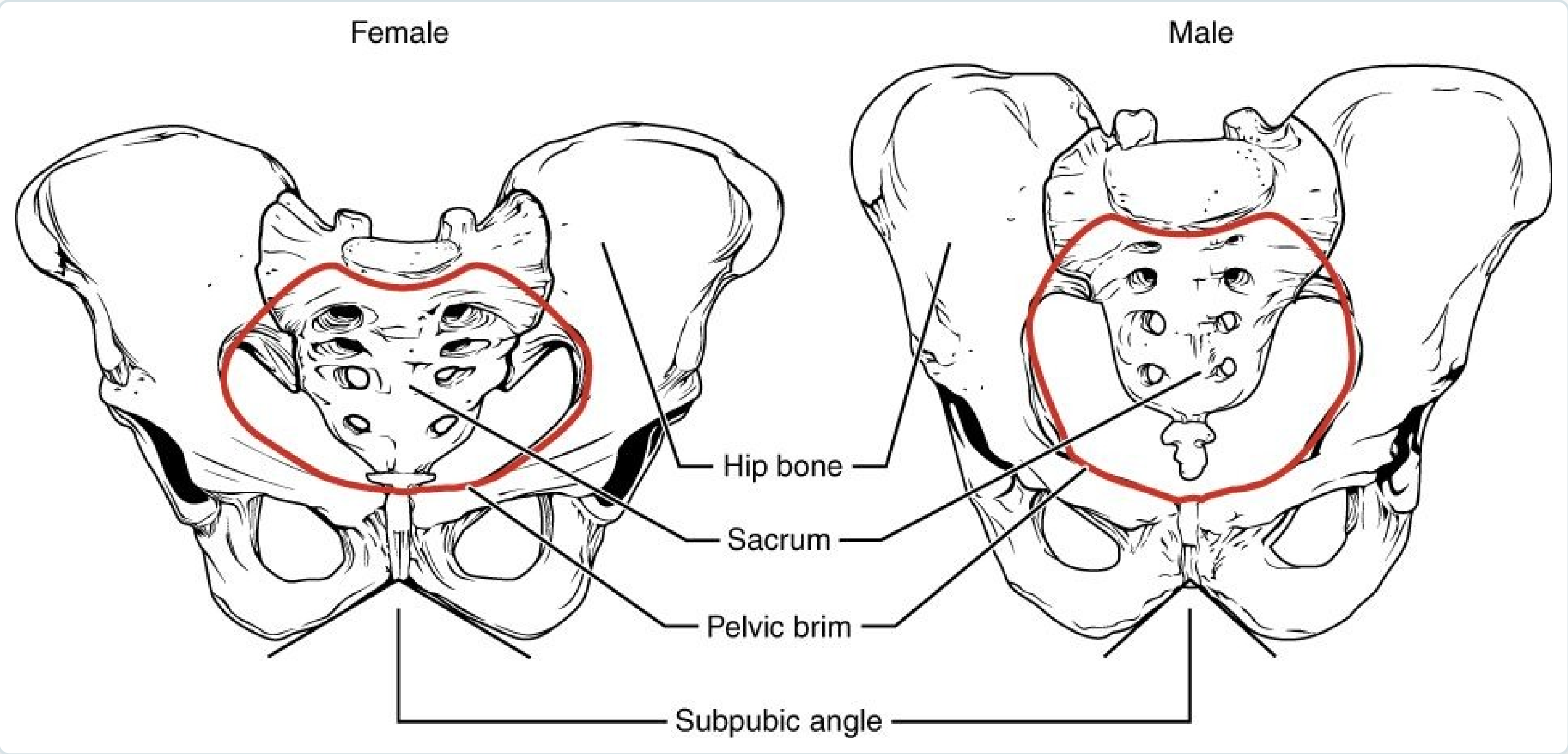
The Hip Bone

- Three fused bones: ilium, ischium, pubis
- Meet at the acetabulum (hip socket)
- Ilium is the broad flaring blade
- Many muscle attachments



The hip bone (os coxae)

Male vs. Female Pelvis



Differences between the male and female pelvis

- Female pelvis is wider and shallower
- Larger outlet for childbirth

- Male pelvis is narrower and heavier
- Sex differences aid identification

8.4

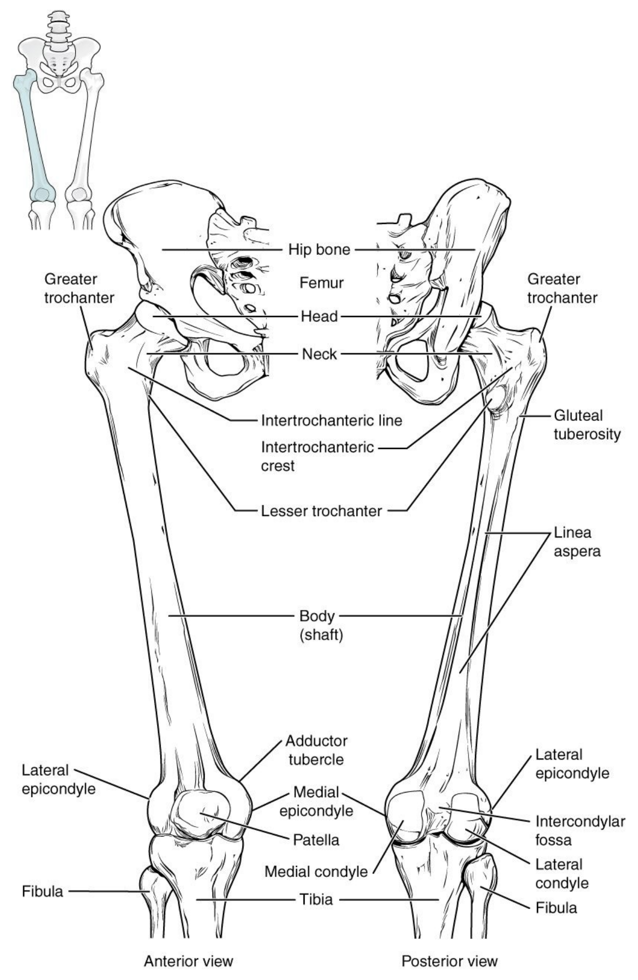
SECTION

Bones of the Lower Limb

The femur, tibia, fibula, and the bones of the foot.

SECTION 8.4

The Femur

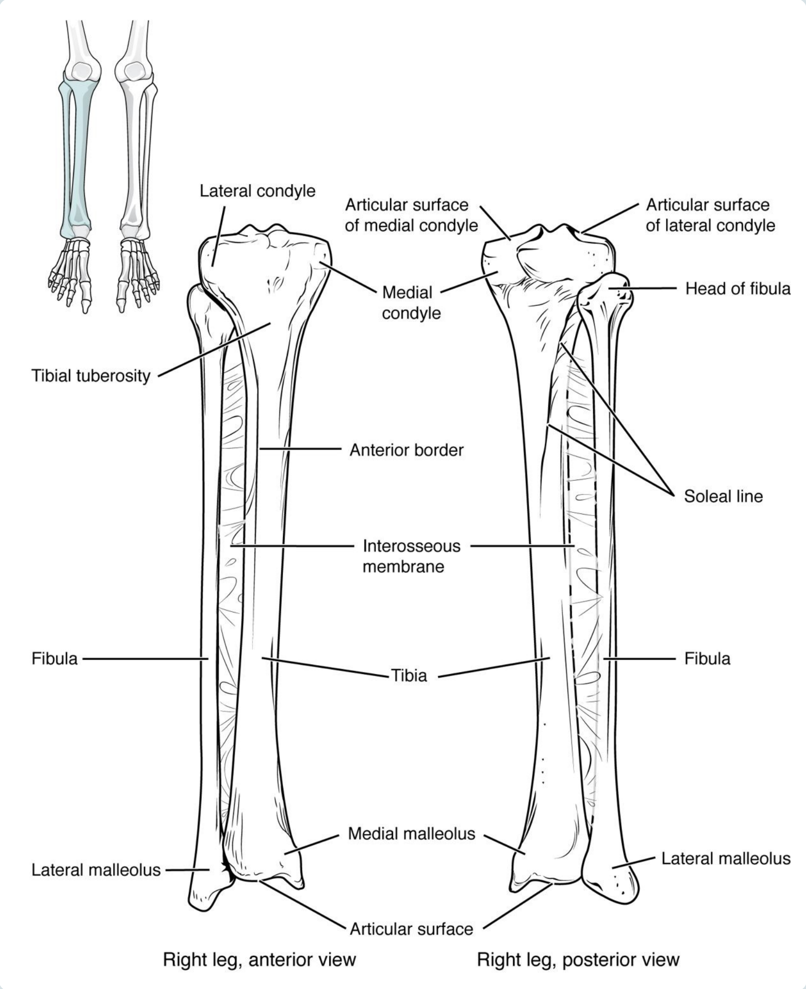


The femur (thigh bone)

- The thigh bone — longest, strongest bone
- Head fits into the acetabulum
- Meets the tibia at the knee
- Bears great mechanical loads

SECTION 8.4

The Tibia and Fibula



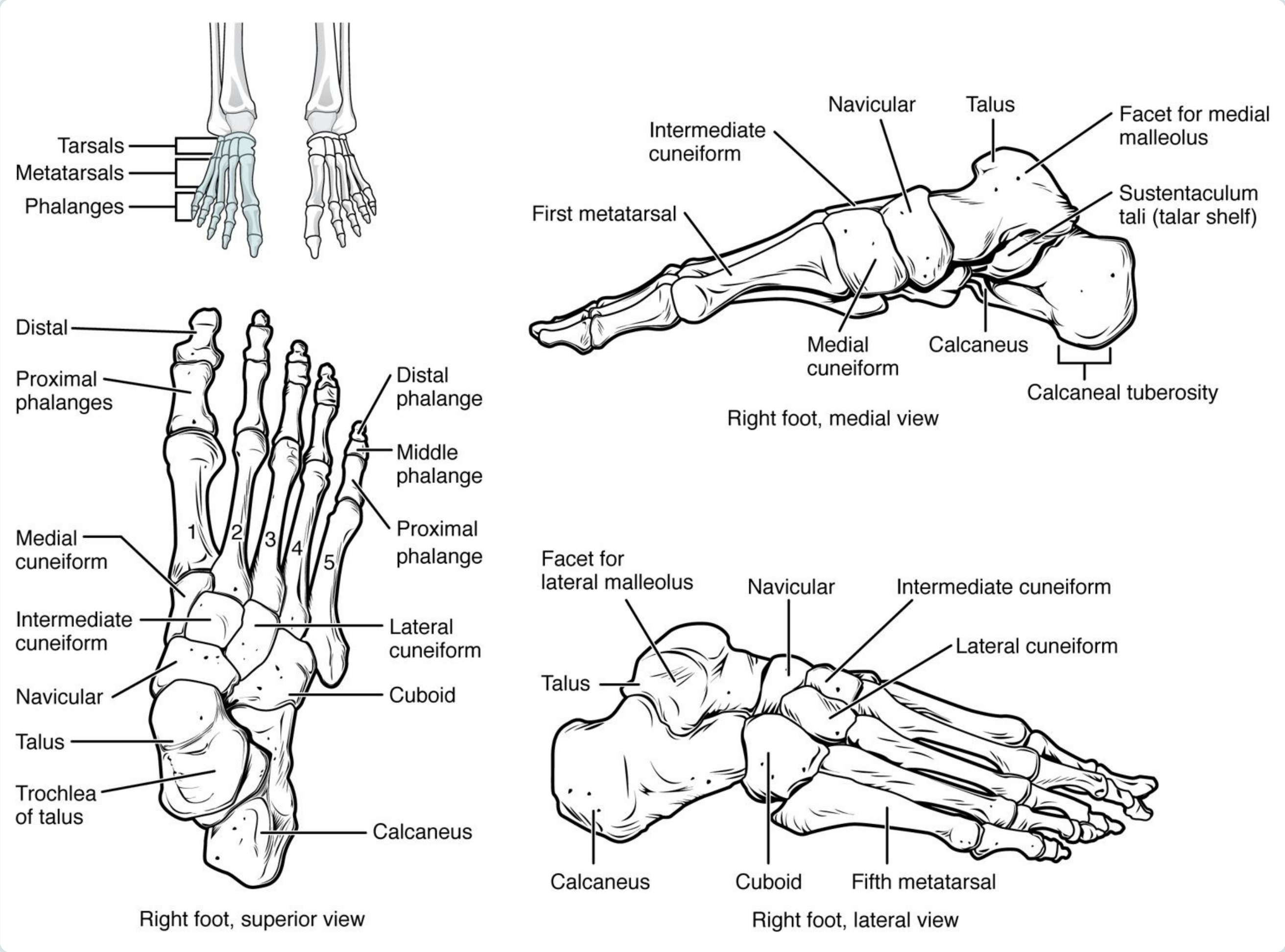
The tibia and fibula (lower leg)

- The two bones of the lower leg
- Tibia (shinbone) bears the weight
- Fibula stabilizes the ankle
- They form the malleoli of the ankle

SECTION 8.4

Bones of the Foot

- 7 tarsals form the ankle and heel
- 5 metatarsals form the sole
- 14 phalanges form the toes
- Arches support and absorb shock



The bones of the foot

8.5

SECTION

Development of the Appendicular Skeleton

How the limbs form before birth, and common congenital differences.

SECTION 8.5

Limb Development & Clubfoot

Clubfoot, a congenital foot condition

- Limbs form early in the embryo
- Bones develop from limb buds
- Clubfoot is a common congenital condition
- Early treatment improves outcomes

CHAPTER 8

Key Takeaways

- The appendicular skeleton is the limbs plus the girdles that attach them.
- The mobile pectoral girdle anchors the arm; humerus, radius, ulna, and hand bones complete the upper limb.
- The stable pelvic girdle bears weight and differs between the sexes.
- The femur, tibia, fibula, and foot bones support the body and enable walking.
- The limbs form early in development; congenital conditions like clubfoot are treatable.

CLINICAL CONNECTIONS

Why the Appendicular Skeleton Matters in Practical Nursing

Fracture Sites

The clavicle, wrist, and hip are common fracture locations.

Carpal Tunnel

Median-nerve compression at the wrist causes numbness and weakness.

Hip Fractures

Femoral-neck fractures are a major risk in older adults.

Obstetrics

The female pelvis shape is central to childbirth.

Mobility

Limb bones support transfers, ambulation, and weight-bearing.

Pediatrics

Congenital limb conditions like clubfoot need early intervention.